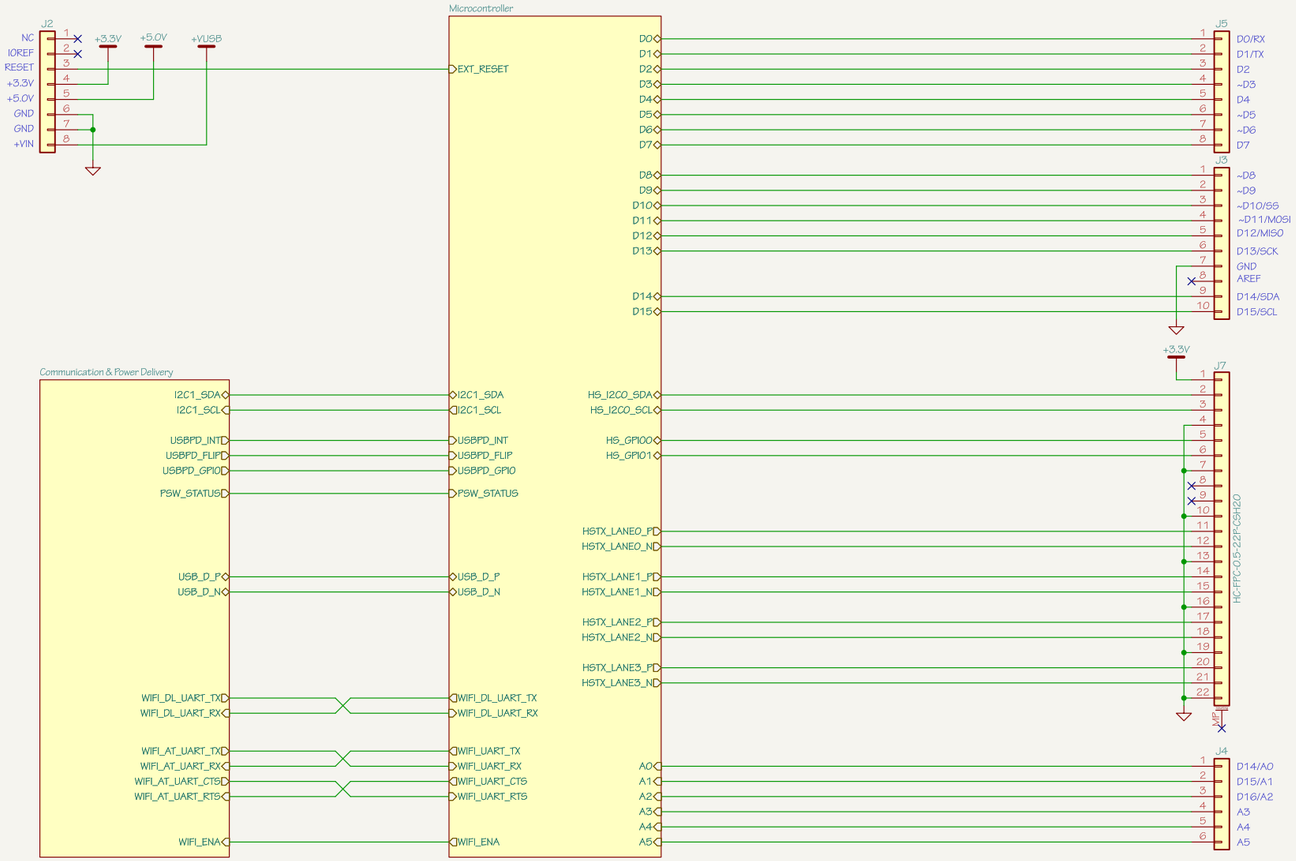


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File: phyduino_pico2.kicad_sch

Title: Phyduino Pico 2

Size: B	Date: 2026-05-04	Rev: A
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Microcontroller

J1
1.0-3PWT
DEBUG CONNECTOR
DEBUG_SWCLK
DEBUG_SWDIO
R1 100Ω
R2 100Ω
TP12
TP15

+3.3V
C20 0.1µF
U4
XC6190AN157R-G
VIN
SW1
SW2
RSTB
VSS
RT
R20 10kΩ
R18 10kΩ
TP19
SYS_nRST
TP26
SW1
1977067-1
C57 0.1µF
EXT_RESET

+3.3V
+3.3VA
+1.1V
U1
RP2350B
RUN
SWCLK
SWDIO
USB_OTP_VDD
QSPI_IOVDD
IOVDD
ADC_AVDD
DVDD
DEBO_SWCLK
DEBO_SWDIO
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCK
QSPI_CS
XIN
XOUT
USB_DM
USB_DP
VREG_AVDD
VREG_VIN
VREG_FB
VREG_LX
VREG_PGND
GND
GPIO100 WIFI_DL_UART_TX
GPIO101 WIFI_DL_UART_RX
GPIO102 ADC_SCK
GPIO103 ADC_MOSI
GPIO104 ADC_MISO
GPIO105 ADC_nCS
GPIO107 WIFI_ENA
GPIO108 WIFI_UART_TX
GPIO109 WIFI_UART_RX
GPIO110 WIFI_UART_CTS
GPIO111 WIFI_UART_RTS
GPIO112 HSTX_LANE0_P
GPIO113 HSTX_LANE0_N
GPIO114 HSTX_LANE1_P
GPIO115 HSTX_LANE1_N
GPIO116 HSTX_LANE2_P
GPIO117 HSTX_LANE2_N
GPIO118 HSTX_LANE3_P
GPIO119 HSTX_LANE3_N
GPIO120 HS_I2C0_SCL
GPIO121 HS_I2C0_SDA
GPIO122 D2
GPIO123 D3
GPIO124 D4
GPIO125 D5
GPIO126 D6
GPIO127 D7
GPIO128 D12 D12/MISO
GPIO129 D10 ~D10/SS
GPIO130 D13 D13/SCK
GPIO131 D11 ~D11/MOSI
GPIO132 D8
GPIO133 D9
GPIO134 D1 D1/UART_TX
GPIO135 D0 D0/UART_RX
GPIO136 D14 D14/I²C_SDA
GPIO137 D15 D15/I²C_SCL
GPIO138 BUILTIN_LED
GPIO139
GPIO140
GPIO141
GPIO142 PSW_STATUS
GPIO143 USBPD_INT
GPIO144 USBPD_GPIO
GPIO145 USBPD_FLIP
GPIO146
GPIO147

U2
GD25LQ128EWIGR
VCC
SI(IO0)
SO(IO1)
WP(IO2)
HOLD(IO3)
SCK
CS
EP
GND
C1 0.1µF
FLASH_SD0
FLASH_SD1
FLASH_SD2
FLASH_SD3
FLASH_CLK
FLASH_CS

SW2
MINI MSK12C02
TP25
R49 4.7kΩ

BOOT SWITCH
- +3.3V Normal Boot mode
- GND UF2 Boot mode

C14 18pF
C15 18pF
Y1 12MHz
R13 4.7kΩ

USB_D_N R3 27Ω
USB_D_P R4 27Ω

R5 27Ω
C4 4.7µF
L3 3.3µH
C3 4.7µF
C2 4.7µF

R46 470Ω
D1 GREEN

External Analog-to-Digital Converter

+3.3V
+3.3V
+5.0V
+5.0VA
R43 4.7kΩ
R42 4.7kΩ
C11 1µF
FB1
C12 1µF
U5
AD57038QRTERQ1
SDO
SDI
SCLK
CS
DECAP
DVDD
AVDD
PAD
GND
AIN0/GPIO100 A0
AIN1/GPIO101 A1
AIN2/GPIO102 A2
AIN3/GPIO103 A3
AIN4/GPIO104 A4
AIN5/GPIO105 A5
AIN6/GPIO106 HS_GPIO0
AIN7/GPIO107 HS_GPIO1
C10 1µF
TP24
TP23
TP20
TP21

Power Indicators & EEPROM

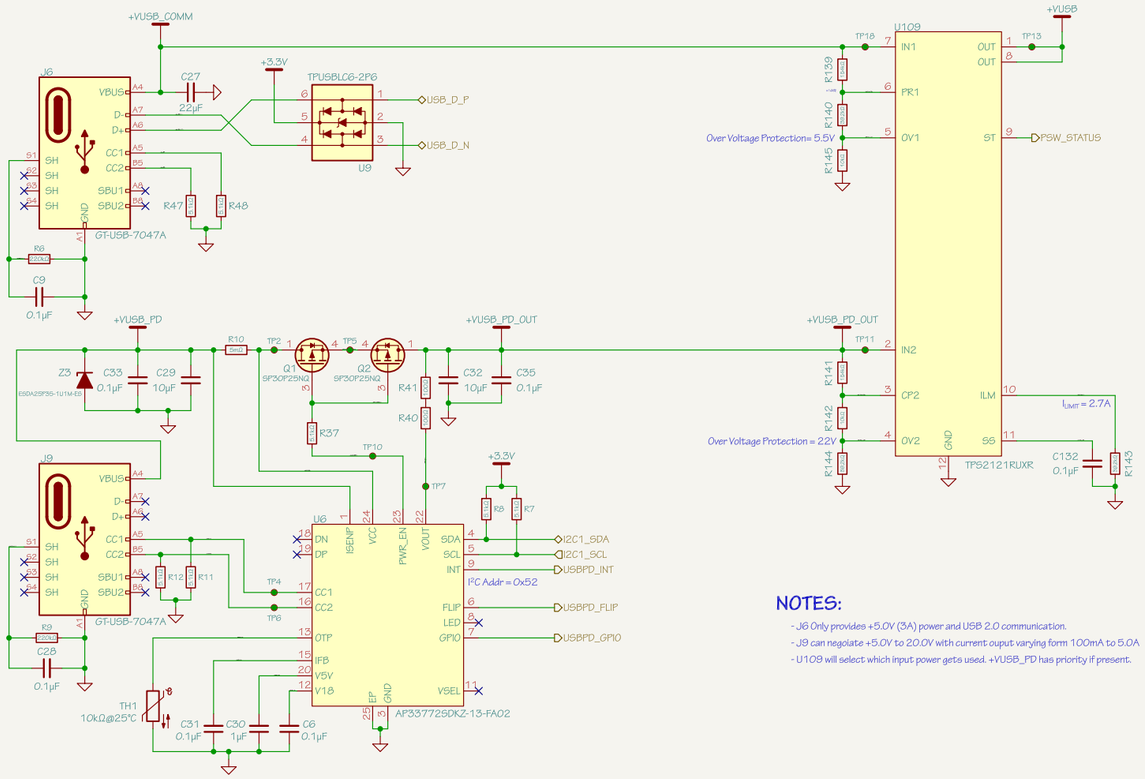
+3.3V
C22 0.1µF
U3
PCA9570GMH
SDA
SCL
I²C Addr = 0x48
P0
P1
P2
P3
RN1A ±1% 330Ω
RN1B ±1% 330Ω
RN1C ±1% 330Ω
RN1D ±1% 330Ω
D5 GREEN
D4 GREEN
D3 GREEN
D2 GREEN
TP1
TP5
I2C1_SDA
I2C1_SCL

+3.3V
C21 0.1µF
U7
ZD24C64A-MAGMT
SDA
SCL
I²C Addr = 0x70
A0
A1
A2
WP
64Kb
EP

+1.1V
C119 0.1µF
C135 0.1µF
C120 4.7µF
+3.3V
C129 0.1µF
C125 0.1µF
C110 0.1µF
C111 0.1µF
C26 0.1µF
C34 0.1µF
C133 0.1µF
C130 0.1µF
C126 0.1µF
FB100
+3.3VA
C117 0.1µF

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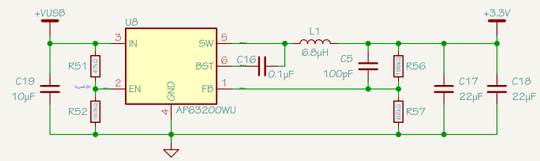
USB Power Delivery



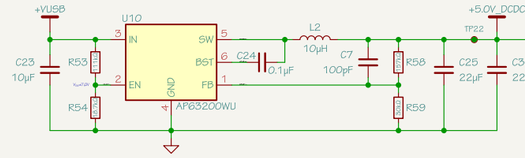
NOTES:

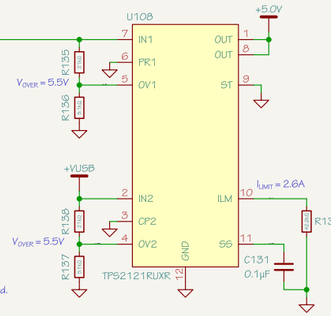
- J6 Only provides +5.0V (3A) power and USB 2.0 communication.
- J9 can negotiate +5.0V to 20.0V with current ouput varying form 1.00mA to 5.0A
- U109 will select which input power gets used. +VUSB_PD has priority if present.

System Power Regulators



U108 : Selects between +VUSB and +5.0V_DCDC. If +5.0V_DCDC is present then it is selected over +VUSB, otherwise +VUSB is selected.

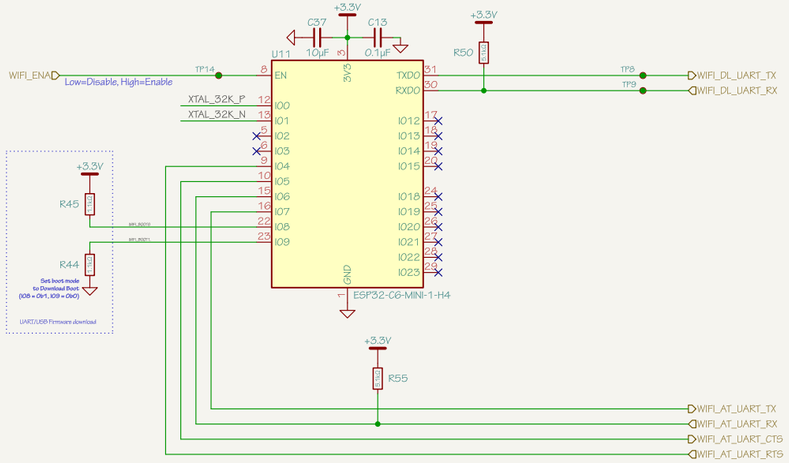




NOTES:

- U8 will only enable when +VUSB is above +4.3V.
- U10 will only enable when +VUSB is above +7.0V.
- U108 select between +VUSB and the output of U10. If both voltages are below +5.5V, U10's output will be selected.

WiFi Module Interface



NOTES:

- To download ESP-AT firmware to ESP32 module use UART0 pins (TXD0 & RXD0).
- Both modules are pin compatiable, either can be used.
- Using ESP-AT as the network interface API.
- GitHub fork repo for ESP-AT: https://github.com/perc-sw/esp-at

ESP32-C61-MINI-1:

- ESP-AT version 5.0.1.0
- Hardware connection guide : https://docs.espressif.com/projects/esp-at/en/latest/esp32c61/Get_Started/Hardware_connection.html

Required Hardware connections

Connection	Pin	Direction
Download/Log RX	RXD0 (GPIO10)	INPUT
Download/Log TX	TXD0 (GPIO11)	OUTPUT
AT command/response RX	IO6 (GPIO6)	INPUT
AT command/response TX	IO7 (GPIO7)	OUTPUT
AT command/response CTS	IO5 (GPIO5)	OUTPUT
AT command/response RTS	IO4 (GPIO6)	INPUT

- 2.4GHz Wi-Fi 6 & BLE 5.0
- **Not for low power applications.**

ESP32-C6-MINI-1:

- ESP-AT version 4.1.1.0
- Hardware connection guide : https://docs.espressif.com/projects/esp-at/en/latest/esp32c6/Get_Started/Hardware_connection.html

Required Hardware connections

Connection	Pin	Direction
Download/Log RX	RXD0 (GPIO17)	INPUT
Download/Log TX	TXD0 (GPIO16)	OUTPUT
AT command/response RX	IO6 (GPIO6)	INPUT
AT command/response TX	IO7 (GPIO7)	OUTPUT
AT command/response CTS	IO5 (GPIO5)	OUTPUT
AT command/response RTS	IO4 (GPIO4)	INPUT

- 2.4GHz Wi-Fi 6 & BLE 5.0

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